kader Academy

Machine Safety Specialist (CE)

Tab 1 – Module 4: Partly completed machinery and modifications

TAB1: principles machine safety
Tab1-4/5 20220812

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Partly completed machinery

The definition of partly completed machinery (MD article 2):
"'partly completed machinery' means an assembly which is almost machinery but which cannot in itself perform a specific application. A drive system is partly completed machinery. Partly completed machinery is only intended to be incorporated into or assembled with other machinery or other partly completed machinery or equipment, thereby forming machinery to which this Directive applies;"

Examples of partly completed machinery:

- Drive system
- Industrial robot

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Tab1-4/5-Sh.3

Partly completed machinery

Guide to application of the Machinery Directive 2006/42/EC

Edition 2.2 – October 2019
(Update of 2nd Edition)

The guide to application of Machinery Directive provides in §46 additional explanation of the definition.

- Partly completed machinery is a product that is similar to machinery.
- In other words an assembly consisting of linked parts or components at least one of which moves, but which lacks some elements necessary to perform its specific application.
- Partly completed machinery must thus undergo further construction in order to become final machinery that can perform its specific application.

Tab1-4/5-Sh.4

Partly completed machinery

Guide to application of the Machinery Directive 2006/42/EC

Edition 2.2 – October 2019
(Update of 2nd Edition)

Industrial robots are mentioned as an example for partly completed machinery because:

- Partly completed machinery must thus undergo further construction in order to become final machinery that can perform its specific application. (only a number of axes possibly with controls but without a specific function);
- The manufacturer of the final machinery takes the necessary measures to make sure that the robot can perform its specific application safely within the assembly.
- In practice, only an industrial "stand and function alone-robot" provided with both an end-effector and control system so that it can itself perform a specific application, is a complete machinery under the Machinery Directive. *(it must be delivered with both the tooling (like a gripper) and a control system so that it can perform a specific application independently)*

Tab1-4/5-Sh.5

Guide to application of the Machinery Directive 2006/42/EC

Edition 2.2 – October 2019
(Update of 2nd Edition)

Partly completed machinery

"Further construction" does not mean:

- the fitting of a drive system to machinery supplied without a drive system…
- the connection on site or with sources of energy….

Partly completed machinery should also be distinguished from machinery ready to be installed on a means of transport or in a building or a structure.

Tab1-4/5-Sh.6

Guide to application of the Machinery Directive 2006/42/EC

Edition 2.2 – October 2019
(Update of 2nd Edition)

Partly completed machinery

Machinery that meets the definition in the first three indents of Article 2(a) so it can in itself operate independently, performing its specific application. But which only lacks the necessary protective means or safety components, such as guards, is not to be considered as partly completed machinery.

Such incomplete machinery does not meet the requirements of the Machinery Directive and must not be CE marked and cannot be placed on the EU/EEA market.

Sometimes machinery missing some parts is confused with a PCM. If the missing parts are not constitutional parts of the machinery and do not give rise to substantial impact on the safety of that machinery, the manufacturer must provide instructions and clear information for the installation of the missing parts, such that the safety and compliance of the final machinery is guaranteed with the Machinery Directive.

Tab1-4/5-Sh.7

Partly completed machinery

EUROPEAN COMMISSION

COMMISSION NOTICE
The 'Blue Guide' on the implementation of EU product rules 2022

(Text with EEA relevance)

(2022/C 247/01)

Related explanation below about the responsibilities of the manufacturer of a machine (product).

Blue Guide 2022 §3.1:

- The manufacturer who subcontracts some or all of his activities may in no circumstances discharge himself from his responsibilities, for example to an authorized representative, a distributor, a user or a subcontractor.

Tab1-4/5-Sh.8

Partly completed machinery – Case 1

- The manufacturer of a conveyor for bulk materials with defined maximum dimensions and weight etc. of the bulk materials to be transported, delivers this conveyor (complete incl. frame) without software and possibly without drive system.
- The software is provided by a system integrator or the customer, the manufacturer can define clear conditions and instructions for assembly, installation and commissioning in the operating instructions.

Is it allowed to supply this conveyor as partly completed machinery?

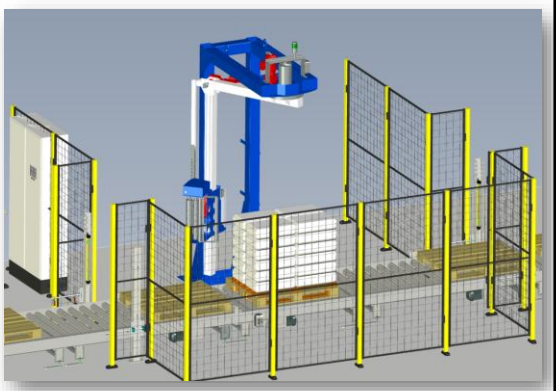
Partly completed machinery – Case 2

- The manufacturer of a conveyor for bulk materials with defined maximum dimensions and weight etc. of the bulk materials to be transported, delivers this conveyor (complete incl. frame)
- Based on the manufacturer's risk assessment, an inspection hatch is required which stops the drive when it is opened (interlocking guard). The customer wants to have the control system and the safety function to be realized by another party (or itself).
- The manufacturer has defined clear conditions and instructions for assembly, installation and commissioning in the operating instructions.

Is it allowed to supply this conveyor as partly completed machinery?

Partly completed machinery – Case 3

- A pallet wrapper is supplied by the manufacturer with a fixed guard for 3 out of 4 sides where the dangerous parts are accessible
- This is because the customer wants to place the machine close to an existing wall. The fixed guard on the wall side is not necessary.
- The manufacturer indicates in the instructions that the machine is explicitly intended to be placed this 'wall arrangement' and sets requirements for attaching the fixed guard to the floor and wall.

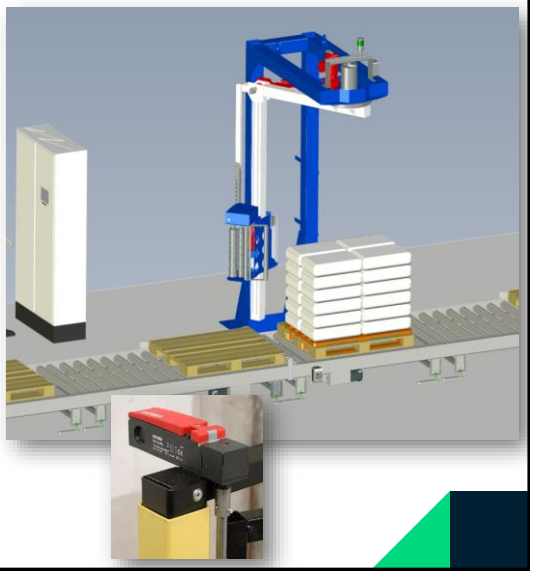


Is it allowed to supply this pallet wrapper as partly completed machinery?

Partly completed machinery – Case 4

- A pallet wrapper is supplied by the manufacturer without the guard.
- This is because the customer wants to install the guarding himself and wants to be able to tailor it to the space where the machine is set up.
- The manufacturer supplies the required sensor for guard locking on the access door, together with an installation instruction.

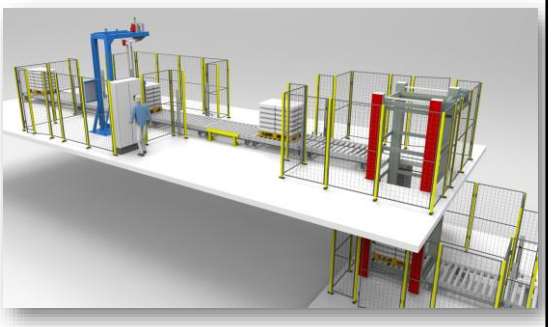
Is it allowed to supply this pallet wrapper as partly completed machinery?



Partly completed machinery – Case 5

- A large machine is assembled and tested in the manufacturer's own production hall (mostly, but not 100% complete).
- The machine is transported in parts to the customer, where it is finished on site by the manufacturer.

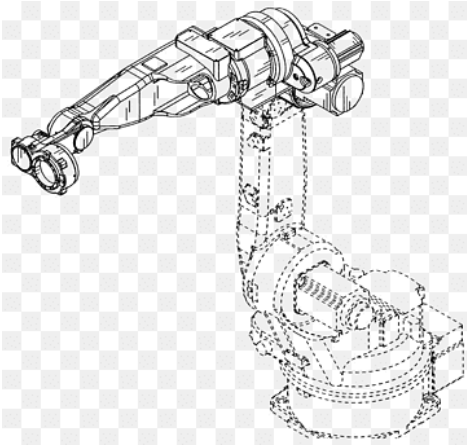
Is it allowed to supply this production line as partly completed machinery?



Partly completed machinery – Case 6

- An industrial robot is supplied without 'robot cell' / guards, protections and specific application.
- The manufacturer provides clear instructions and conditions to allow the robot to move safely.

Is it allowed to supply this industrial robot as partly completed machinery?



When does II.1.B apply?

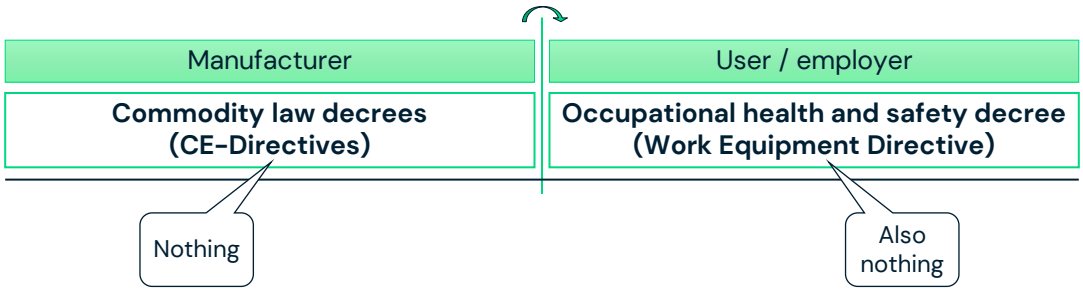
Only if the manufacturer cannot foresee or cover certain risks, because…

- certain information is not available from the client,
- this cannot be covered with instructions either,
- there is no question of a "specific application".

This should correspond to not being able to meet certain requirements from Annex I of the Machinery Directive, for the above or similar reasons.

Modifying existing machinery

What does the (current) legislation say about modifications and CE conformity?



The successor to the Machinery Directive (Machine Regulation) will include a definition of a 'substantial modification'.

Modifying existing machinery

The term 'substantial modification'

Derived from the guide to the application of the Machinery Directive

- [§72] 'The question arises as to when a transformation of machinery is considered as construction of new machinery subject to the Machinery Directive.'
- [§82] 'On the other hand, if the modification is substantial (for example, a change of function and/or performance of the machinery) and not foreseen or agreed by the manufacturer… '

Modifying existing machinery

The term 'important change'

Addition Blue Guide 2022

- [§2.1] A product which has been subject to important changes or overhauls after it has been put into use, may be considered as a new product when:

  i) its original performance, purpose or type is modified, without this being foreseen in the initial risk assessment;

  ii) the nature of the hazard has changed or the level of risk has increased in relation to the relevant Union harmonization legislation; and

  iii) the product is made available (or put into service if the applicable legislation also covers putting into service within its scope). This has to be assessed on a case-by-case basis and, in particular, in view of the objective of the legislation and the type of products covered by the legislation in question.

Modifying existing machinery

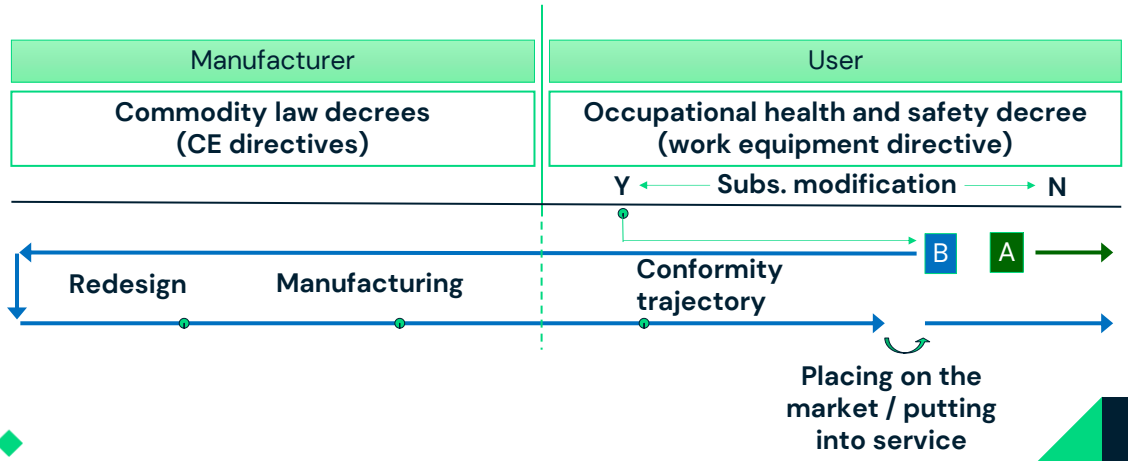
Changes and 'explanatory notes on the law'

- Explanation on the European product directives: 'Blue Guide'.
- Explanation on the Machinery Directive: 'Guide to the application of the Machinery Directive'.
  a. For 'Substantial Modification' (SM): consider 'as new'.
  b. Various reservations: 'assess on a case-by-case basis'.
  c. No clear criteria, no examples.
  d. Risk assessment is the basis for the choice of SM or not.
  e. The 'change manager' determines this choice.
  f. For SM: conformity with all current directives required.

Additional information from the guide:
When replacing a subassembly: no CE process for the entire line, provided there are limited safety influences (integration assessment).

Modifying existing machinery

Schematic view:



Modifying existing machinery

Werkinstructie
Beoordelen van gewijzigde machines

NLA published a work instruction "Assessment of Modified Machines" in 2022.

"....The experiences over the past years and the developments regarding the new Machinery Regulation have prompted the Labour Inspectorate to establish a work instruction on how to deal with modifications to machines."

"This instruction provides guidance on current laws and regulations and is pragmatic, future-proof, and proportional. The work instruction addresses machines manufactured before 1995 that fall under the Machinery Directive 2009/104/EC, machines covered by the current Machinery Directive 2006/42/EC, and machines covered by the Machinery Regulation..."

Modifying existing machinery

Werkinstructie
Beoordelen van gewijzigde machines

Determining whether a substantial modification has occurred:

- *"The person modifying or altering a machine must assess whether this modification constitutes a 'substantial modification.'*
- *"For this purpose, the modifier must at least conduct a risk assessment as described in the 'General Principles' of Annex I of the Machinery Directive 2006/42/EC.*
- *"If the modifier determines that a substantial modification has occurred, they bear the full manufacturer's obligations for the entire machine."*

Modifying existing machinery

Werkinstructie
Beoordelen van gewijzigde machines

What to do in case of doubt or uncertainty?

"In situations where it is unclear whether the modification constitutes a 'substantial modification,' there exists a German document titled 'Wesentliche Veränderung von Maschinen' (Substantial Modification of Machinery). This document provides interpretation by the German regulatory authority on what constitutes a substantial modification. This interpretation is used across Europe to assess whether a modification to a machine is substantial or not.

"If there is still doubt regarding whether a 'substantial modification' has occurred, the guide (§72) for the application of the Machinery Directive recommends consulting the national authorities. The Safety and Products department (VP) of the Inspection-wide Knowledge Center (IKC) of the Dutch Labour Inspectorate answers questions regarding this matter."

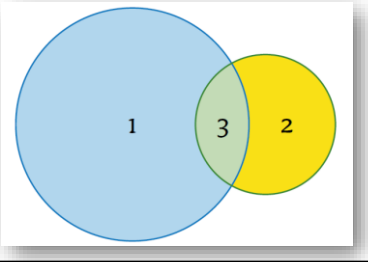
Modifying existing machinery

Werkinstructie
Beoordelen van gewijzigde machines

he work instruction distinguishes between the following scenarios:

- Route 1: 'Substantially modifying' an existing CE-marked machine.
- Route 2: ' Substantially modifying' an existing machine without a CE marking (from before 1995).

The proposed approach to pragmatically address these two routes is detailed in the work instruction using the following graphical representation of the existing machine and the modification:



Modifying existing machinery

Werkinstructie
Beoordelen van gewijzigde machines

Route 1: 'Substantially modifying' an existing CE-marked machine:

1. To substantiate the Essential Health and Safety Requirements (EHSRs) in accordance with Annex I of the Machinery Directive 2006/42/EC, one can utilize the original documentation (manual and EC Declaration of Conformity) for the part unaffected by the modification.

2. The person performing the modification must substantiate that the new part fully complies with the requirements of the Machinery Directive.

3. The influences of the modification on the base machine must be fully substantiated as described in point 2 above. The boundaries of the modification affecting the base machine must be clear.

Modifying existing machinery

Werkinstructie
Beoordelen van gewijzigde machines

Route 2: 'Substantially modifying' an existing machine without a CE marking (from before 1995)

1. This part of the base machine must comply with the minimum requirements for safety and health during the use of equipment by workers in the workplace, as stipulated in the Work Equipment Directive 2009/104/EC.

2. The person making the modification must substantiate that the new part fully complies with the requirements of the Machinery Directive.

3. The influences of the modification on the base machine must be fully substantiated as described in point 2 above. The boundaries of the modification affecting the base machine must be clear.

Modifying existing machinery

Werkinstructie
Beoordelen van gewijzigde machines

Route 2: 'Substantially modifying' an existing machine without a CE marking (from before 1995)

In addition to point 3 of the previous sheet, the following applies:

a) There is no new hazard or no increase in an existing hazard, so the machine can still be considered safe.

b) Although there is a new hazard or an increase in an existing hazard, the existing protective measures of the machine from before the modification are still adequate. In this case, the machine can still be considered safe.

c) There is a new hazard or an increase in an existing hazard, and the existing protective measures are not adequate or suitable. This means that for this part, the machine must be adapted, and this adaptation must comply with Directive 2006/42/EG.

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